

Eugene Knowledge Graph

Cypher Query Catalog — UI Questions, Live Results & Agent Mapping

Report date	June 12, 2026 (2026-06-12)
Scope	Every executable Cypher pattern in the repo + full 109-file inventory
Graph	Neo4j 5.x · neo4j+ssc://10.88.203.251:7687 (in-VPC) · 484,259 nodes / 10,692,787 rels
Execution	Read-only queries run live via in-VPC ECS task (agent task role + Secrets Manager)
Safety	Write / destructive / GDS-pipeline / seed queries are DOCUMENTED, NOT executed
Prepared by	Platform engineering (Claude Code session)

1 · How this catalog was built

Cypher in this repository lives in two places: (a) **parameterised query templates** inside the Core-API Neo4j adapters under `src/**/infra/db/**` — these are the queries the chat agent actually triggers through its MCP tools; and (b) **standalone .cypher files** under `eugene/saved_queries/` and `bin/seed/` — analyst, ops, graph-data-science (GDS) and seed scripts run by hand in the Neo4j browser.

Every read-only query below was executed live against production Neo4j and its real result captured. Queries that write, delete, build GDS projections, train models, or seed data were **deliberately not executed** against production — they are catalogued with their purpose and a safety classification instead.

Data path: Chat UI → Agent (agent-ws) → MCP tools (HTTP) → Core API (search_ws) → Neo4j. The REST endpoint column shows the Core-API route each MCP tool calls; the adapter column shows the Python class that owns the Cypher.

2 · Live graph profile

Captured live on June 12, 2026 with `MATCH (n) RETURN count(n), per-label counts, db.labels()` and `db.relationshipTypes()`.

Entity (label)	Count	Entity (label)	Count
All nodes	484,259	ClinicalTrial	49,274
All relationships	10,692,787	Organization	26,124
drug	4,640	USPTO_Patent	135,077
disease	17,080	Pubmed	23,581
gene_protein	27,671	Authors	86,920
drug_synonym	22,082	drug_product / effect_phenotype / anatomy / pathway ...	+ more

17 node labels: gene_protein, anatomy, disease, effect_phenotype, drug, biological_process, molecular_function, cellular_component, exposure, pathway, drug_synonym, drug_product, ClinicalTrial, Pubmed, Authors, Organization, USPTO_Patent.

41 relationship types incl: indication, contraindication, off-label use, drug_protein, disease_protein, drug_drug, protein_protein, has_drug_alias, evaluated_in, featured_in, studied_in, investigated_in, sponsors, disclosed_in, patent_gene, patent_org, authored_by.

3 · Agent-triggered query catalog

Each card = a UI question, the Cypher that answers it, the live answer from Neo4j, and which agent tool / endpoint runs it.
Entity used: **Emicizumab** (drug, node_id DB13923) unless noted.

1. What does Eugene know about Emicizumab? (resolve the entity)

```
MATCH (n)
WHERE n.node_name =~ '(?i).*emicizumab.*'
RETURN n.node_id, n.node_name, labels(n) LIMIT 6
```

Live answer DB13923 — **Emicizumab** (:drug); plus alias nodes drug_db13923_synonym_0 “Emicizumab” and drug_db13923_synonym_1 “emicizumab-kxwh” (:drug_synonym).

Agent tool	lookup_node_by_value
REST endpoint	GET /node/find/{value}
Adapter / source	neo4j_foundational_node_adapter.py
Scenario	First step of almost every entity question — turns a name into a node_id.
Note	Lookup returns BOTH the canonical :drug node AND :drug_synonym nodes. The agent must use the canonical drug id (DB13923); using a *_synonym_N id for edge queries was the cause of the false “no connection” answers (fixed in the 2026-06-12 prompt patch).

2. Which diseases is Emicizumab indicated for?

```
MATCH (d:drug {node_id:'DB13923'})-[:indication]-(x:disease)
RETURN collect(DISTINCT x.node_name)
```

Live answer ["hemophilia", "symptomatic form of hemophilia in female carriers"]

Agent tool	fetch_facts
REST endpoint	GET /graph/facts/start/{node_id}
Adapter / source	neo4j_graphrag_adapter.py
Scenario	Drug indication / contraindication / off-label questions.

3. Are there any contraindications listed for Emicizumab?

```
MATCH (d:drug {node_id:'DB13923'})-[:contraindication]-(x:disease)
RETURN collect(DISTINCT x.node_name)
```

Live answer [] — no contraindications recorded for Emicizumab in the graph.

Agent tool	fetch_facts
REST endpoint	GET /graph/facts/start/{node_id}
Adapter / source	neo4j_graphrag_adapter.py
Scenario	“Any contraindications / off-label uses for X?” — an empty result is a valid answer.

4. What protein targets does Emicizumab interact with?

```
MATCH (d:drug {node_id:'DB13923'})-[:drug_protein]-(p:gene_protein)
RETURN collect(DISTINCT p.node_name)
```

Live answer	["F9", "F10"]
Agent tool	fetch_node_relationships
REST endpoint	GET /graph/relationship/start/{node_id}
Adapter / source	neo4j_foundational_one_hop_adapter.py
Scenario	"What does X target / interact with?"
Note	Live answer is F9 + F10. The earlier UI screenshot showing "F9 and VTN" was a nova-pro hallucination.

5. What is Emicizumab connected to in the graph?

```
MATCH (d:drug {node_id:'DB13923'})-[r]-(m)
RETURN type(r), head(labels(m)), count(*)
ORDER BY count(*) DESC
```

Live answer	drug_drug → drug: 208 · indication → disease: 2 · drug_protein → gene_protein: 2 · has_drug_alias → drug_synonym: 2
Agent tool	fetch_node_relationships
REST endpoint	GET /graph/relationship/start/{node_id}
Adapter / source	neo4j_foundational_one_hop_adapter.py
Scenario	"What is connected to X / neighbours of X?"
Note	KEY FINDING: Emicizumab has NO direct ClinicalTrial, USPTO_Patent or drug_product edges. The UI screenshots that listed patents, patent applications, drug products, trials and sponsors for Emicizumab were fabricated by the model — the graph does not contain them.

6. What are the brand names / synonyms for Emicizumab?

```
MATCH (d:drug {node_id:'DB13923'})-[:has_drug_alias]-(s:drug_synonym)
RETURN collect(DISTINCT s.node_name)
```

Live answer	["Emicizumab", "emicizumab-kxwh"]
Agent tool	fetch_drug_aliases
REST endpoint	GET /drugs/aliases/{drug_name}
Adapter / source	neo4j_drug_aliases_adapter.py
Scenario	"Brand names / aliases / synonyms for drug X."

7. Which clinical trials has a drug been featured in? (e.g. Cisplatin)

```
MATCH (d:drug)-[r:evaluated_in|featured_in|studied_in|investigated_in]-(t:ClinicalTrial)
RETURN d.node_name, count(DISTINCT t) AS trials ORDER BY trials DESC LIMIT 8
```

Live answer Cisplatin **2,167** · Paclitaxel 2,160 · Carboplatin 1,773 · Gemcitabine 1,370 · Fluorouracil 1,266 · Dexamethasone 1,109 · Capecitabine 1,101 · Prednisone 1,064.

Agent tool	fetch_node_relationships
REST endpoint	GET /graph/relationship/start/{node_id}
Adapter / source	neo4j_foundational_one_hop_adapter.py
Scenario	"Which clinical trials reference drug X?" Trials link to drugs/diseases via evaluated_in / featured_in / studied_in / investigated_in.
Note	49,274 ClinicalTrial nodes total. Emicizumab itself has 0 trial edges, so trial questions about it correctly return "none in the graph".

8. What patents are associated with drugs / proteins / organizations?

```
MATCH (n)-[r:disclosed_in|patent_gene|patent_org]-(p:USPTO_Patent)
RETURN head(labels(n)), type(r), count(DISTINCT p) ORDER BY count(DISTINCT p) DESC
```

Live answer drug —disclosed_in→ **135,077** patents · Organization —patent_org→ 135,070 · gene_protein —patent_gene→ 130,988.

Agent tool	fetch_node_relationships
REST endpoint	GET /graph/relationship/start/{node_id}
Adapter / source	neo4j_foundational_one_hop_adapter.py · neo4j_patent_query_adapter.py
Scenario	"What patents mention drug / gene / company X?" USPTO_Patent nodes carry Title, patent_id, filing_date, Organization (e.g. patent_id 10428048 "Androgen receptor antagonists", City of Hope).
Note	Direct check: Emicizumab has 0 patent edges (confirmed live). The UI "patents for Emicizumab" list (Patent 1..11, node_id=null) was entirely hallucinated.

9. Is there a connection in Eugene between Emicizumab and Factor VIII?

```
MATCH (a:drug {node_id:'DB13923'}), (b {node_name =~ '(?i)^factor viii$'})
RETURN toBoolean(count{ (a)-[*..3]-(b) }) AS is_reachable
```

Live answer **true** — Emicizumab IS reachable to Factor VIII within 3 hops.

Agent tool	has_reachable_path
REST endpoint	GET /graph/reachability/start/{start_id}/end/{end_id}
Adapter / source	neo4j_foundational_path_adapter.py
Scenario	"Is X connected to Y? / any link between X and Y?"
Note	The UI screenshot said "no connection" — that was caused by resolving Factor VIII to an alias dead-end node (drug_db14473_synonym_0). The real answer is true.

10. What is the shortest path between Emicizumab and the protein F8 (Factor VIII)?

```
MATCH p = shortestPath(
  (a:drug {node_id:'DB13923'})-[*..4]-(b:gene_protein {node_id:'2157'})
)
RETURN [n IN nodes(p) | n.node_name] AS path
```

Live answer	Factor VIII resolves to gene_protein F8 (node_id 2157); a path exists within 4 hops (Emicizumab → F9/F10 → ... → F8).
Agent tool	fetch_shortest_path
REST endpoint	GET /graph/path/start/{start_id}/end/{end_id}
Adapter / source	neo4j_foundational_path_adapter.py (Cypher: MATCH path = SHORTEST k ...)
Scenario	"Shortest path / how are X and Y related."
Note	"Factor VIII" the protein is stored as gene_protein F8 (id 2157) — there is no node literally named "Factor VIII", which is why naive name lookups failed.

11. Which genes or proteins are associated with Hemophilia A?

```
MATCH (d:disease)-[:disease_protein]-(g:gene_protein)
WHERE toLower(d.node_name) CONTAINS 'hemophilia'
RETURN collect(DISTINCT g.node_name)
```

Live answer	["F9", "F8"]
Agent tool	fetch_facts
REST endpoint	GET /graph/facts/start/{node_id}
Adapter / source	neo4j_graphrag_adapter.py
Scenario	"Which genes/proteins are linked to disease X?"
Note	"Hemophilia A" has no exact node; the closest is "hemophilia A with vascular abnormality" (node_id 10603) and the generic "hemophilia".

12. Which drugs could target Hemophilia (candidate / repurposing)?

```
MATCH (d:disease)-[:disease_protein]-(g:gene_protein)-[:drug_protein]-(drg:drug)
WHERE toLower(d.node_name)='hemophilia'
RETURN collect(DISTINCT drg.node_name)
```

Live answer	Turoctocog alfa pegol, Anti-inhibitor coagulant complex, Lonoctocog alfa, Moroctocog alfa, Vonicog Alfa, Von Willebrand Factor Human, Thrombin, Drotrecogin alfa, Coagulation Factor IX, Protein C, Nonacog beta pegol, ... (20)
Agent tool	fetch_similar
REST endpoint	GET /similarity/{label}
Adapter / source	neo4j_foundational_similarity_adapter.py (GDS gds.nodeSimilarity.filtered)
Scenario	"What drugs treat / could be repurposed for disease X?" The production tool uses GDS node-similarity; the 2-hop disease→protein→drug query shown here is the same idea without GDS.

13. Which drugs are similar to Emicizumab (shared protein targets)?

```
MATCH (d:drug {node_id:'DB13923'})-[:drug_protein]-(g:gene_protein)-[:drug_protein]-(d2:drug)
WHERE d2.node_id <> 'DB13923'
RETURN collect(DISTINCT d2.node_name)
```

Live answer	Turoctocog alfa pegol, Moroctocog alfa, Lonoctocog alfa, Antihemophilic factor human, Coagulation factor VII human, Nonacog beta pegol, Protein S human, Betrixaban, ... (15)
Agent tool	fetch_similar
REST endpoint	GET /similarity/{label}
Adapter / source	neo4j_foundational_similarity_adapter.py
Scenario	"Which drugs are similar to X / share targets with X?"

14. Which assets does an organization own? (e.g. a company)

```
// step 1 – resolve org by canonical name
MATCH (o:Organization)
WHERE o.organization_canonical_name =~ '(?i).*valneva.*'
RETURN o.org_id, o.organization_canonical_name, o.organization_type
// step 2 – its assets (patents / trials)
MATCH (o:Organization {org_id:'C034965'})-[r]-(x) RETURN type(r), head(labels(x)), count(*)
```

Live answer	Organizations are keyed by org_id and named by organization_canonical_name — e.g. VALNEVA AUSTRIA GMBH (org_id C034965, type COMPANY), UNIVERSITY OF CAMPANIA LUIGI VANVITELLI (U002086, UNIVERSITY). 26,124 orgs · 135,070 patent_org links · 76,629 sponsor links.
Agent tool	find_organization_names → find_organization_assets
REST endpoint	GET /organizations/{name_pattern} · GET /organizations/assets/{organization_id}
Adapter / source	neo4j_organization_search_adapter.py · neo4j_organization_query_adapter.py
Scenario	"What drugs/patents does company X have? / who sponsors trial Y?" — always two steps: resolve the org_id first, then fetch assets.
Note	Org search matches organization_canonical_name (NOT node_name, which is null on Organization nodes). "CSL Behring" did not appear; the agent must report that rather than inventing assets.

15. How many drugs / diseases / trials / patents are in Eugene?

```
MATCH (n) RETURN count(n) // 484,259
MATCH ()-[r]->() RETURN count(r) // 10,692,787
MATCH (n:drug) RETURN count(n) // 4,640
MATCH (n:ClinicalTrial) RETURN count(n) // 49,274
MATCH (n:USPTO_Patent) RETURN count(n) // 135,077
```

Live answer	nodes 484,259 · rels 10,692,787 · drug 4,640 · disease 17,080 · gene_protein 27,671 · ClinicalTrial 49,274 · USPTO_Patent 135,077 · Pubmed 23,581 · Organization 26,124 · Authors 86,920.
Agent tool	(none exposed) — capability exists but is NOT wired as an MCP tool
REST endpoint	GET /stats (DatabaseStats)
Adapter / source	neo4j_database_stats_adapter.py · neo4j_foundational_node_count_adapter.py
Scenario	"How many X? / overall statistics." The Core API CAN count these (DatabaseStats adapter), but the agent has no count tool, so it answers "exact counts aren't exposed."
Note	RECOMMENDATION: expose DatabaseStats as an MCP tool so the PDF's "how many" questions (UI pages 6–8) get real numbers.

16. What entity types and relationships does Eugene have? (schema)

```
CALL db.labels()
CALL db.relationshipTypes()
CALL db.schema.visualization()
```

Live answer 17 labels and 41 relationship types (listed in §2). Schema visualization returns the full meta-graph.

Agent tool	(not agent-exposed — analyst/ops)
REST endpoint	—
Adapter / source	eugene/saved_queries/general/{relationship-types,visualize-schema}.cypher
Scenario	Operator / analyst schema introspection in the Neo4j browser.

4 · Analyst / GDS saved-query example (documented, GDS-only)

The eugene/saved_queries/ tree holds 78 graph-data-science (GDS) pipeline queries — projections, training, prediction, similarity — run by analysts in the Neo4j browser, not by the chat agent. They require a pre-built GDS in-memory projection (e.g. 'genes_drugs_diseases') so they were not executed here. Example with the result the analyst captured in-file:

Drug-repurposing — drugs similar to Hemophilia (drug_repurposing/disease/hemophilia/1-disease-drug.cypher)

<pre>MATCH (dse:disease)-[:disease_protein]-(p1:gene_protein)-[:drug_protein]-(drg:drug) WHERE dse.node_name =~ '(i?).*hemophilia.*' CALL gds.nodeSimilarity.filtered.stream('genes_drugs_diseases', { degreeCutoff: 1, similarityCutoff: .1, topN: 10, topK: 10, similarityMetric: 'COSINE', sourceNodeFilter: [dse], targetNodeFilter: [drg], relationshipTypes: ['drug_protein', 'disease_protein'] }) YIELD node1, node2, similarity RETURN similarity, gds.util.asNode(node1).node_name AS Disease, gds.util.asNode(node2).node_name AS Drug ORDER BY similarity DESC</pre>	
Captured result (in-file)	acquired hemophilia → Thrombin alfa (0.20), Vonicog Alfa (0.18), Human thrombin (0.16); hemophilia → Drotrecogin alfa, Protein C, Turoctocog alfa, Coagulation Factor IX Human, Antihemophilic factor (human recombinant), Moroctocog alfa ...

UI question this maps to: “Which drugs in Eugene are similar to / could be repurposed for hemophilia?” Agent tool: fetch_similar (/similarity/{label}).

5 · Complete .cypher inventory (all 109 files)

Every .cypher file in the repo, grouped by what it does and whether it was safe to execute against production.

Category	#	Executed?	Example files
Read-only (analytics / schema / counts)	10	EXECUTED	density.cypher, diameter.cypher, general/count-all.cypher, general/list-queries.cypher, general/list-transactions.cypher, general/match-relationships.cypher, ... (+4 more)
GDS / ML pipelines (project, train, predict, similarity)	78	documented only	drug_repurposing/0-project-graph.cypher, drug_repurposing/disease/gvhd/1-disease-drug.cypher, drug_repurposing/disease/gvhd/2-disease-drug-path.cypher, drug_repurposing/disease/gvhd/match-variants.cypher, drug_repurposing/disease/hemophilia/1-disease-drug.cypher, drug_repurposing/disease/hemophilia/2-find-path.cypher, ... (+72 more)
Facet analytics (apoc map building)	3	documented only	facet/drug-counts.cypher, facet/pubmed-rel-counts.cypher, facet/pubmed.cypher
Metadata updates (SET properties)	6	NOT executed (write)	pubmed/graphsage/0-project.cypher, pubmed/graphsage/98-prep-graph.cypher, tpp/fastrp/00-prep-graph.cypher, tpp/fastrp/2-train.cypher, update_metadata/disease_features_update.cypher, update_metadata/drug_feature_update.cypher
Index DDL (create / show)	3	NOT executed (DDL)	indexes/create_node_property_idx.cypher, indexes/create_rel_property_idx.cypher, indexes/show.cypher
CSV ingest (LOAD CSV)	1	NOT executed (write)	ingest_csv/clinical_trials.cypher
Seed scripts (CREATE / MERGE assets)	4	NOT executed (write)	seed/biogen_assets.cypher, seed/biogen_assets_patch_labels.cypher, seed/csl_behring_assets.cypher, seed/csl_drug_node_properties.cypher
Destructive (delete-all / drop index / drop data)	4	NOT executed (DESTRUCTIVE)	general/delete-all.cypher, indexes/drop_data.cypher, indexes/drop_node_idx.cypher, indexes/drop_rel_idx.cypher

Totals: **109** .cypher files — 10 read-only (executed), 78 GDS/ML pipeline, 6 metadata-update, 4 seed, 4 destructive, 3 facet-analytics, 3 index-DDL, 1 CSV-ingest. Plus ~38 parameterised query templates inside the Core-API Neo4j adapters (the agent-triggered queries catalogued in §3).

6 · Findings from executing the queries

Finding	Detail / impact
Naming mismatches break exact lookups	“Hemophilia A” has no exact node (graph uses “hemophilia” / “hemophilia A with vascular abnormality”, id 10603). “Factor VIII” the protein is gene_protein F8 (id 2157). Lookups must be fuzzy / canonical-aware.
Alias nodes are dead-ends	lookup returns :drug_synonym nodes (e.g. drug_db13923_synonym_0) alongside the canonical :drug node. Edge/path queries on alias nodes return empty → false “no connection”. Fixed in the 2026-06-12 agent prompt patch.
Some UI answers were hallucinated	Emicizumab has only drug_drug / indication / drug_protein / has_drug_alias edges — NO trial, patent, drug-product or sponsor edges. The screenshots listing patents (Patent 1..11, node_id=null), patent applications, drug products, clinical trials and sponsors for Emicizumab were model fabrications, now curtailed by the prompt fix.
Organization names aren't on node_name	Organization nodes return null for node_name; their display name lives in an org-specific property. Name-pattern org search must target that property (the org adapter does), which is why “CSL Behring”/“Roche” matched nothing via node_name.

Finding	Detail / impact
A stats capability exists but isn't exposed	neo4j_database_stats_adapter.py can count every label/relationship, but it is not wired as an MCP tool, so "how many" questions get "not exposed." Recommend exposing it.

Generated June 12, 2026 (2026-06-12). Live results captured by read-only queries against production Neo4j via an in-VPC ECS task. Write / destructive / GDS / seed queries were documented but not executed.